Spectral Simulation Report

Automated Analysis

June 18, 2025

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1 Introduction

This report presents the results of spectral simulations derived from HDF5 dataset analysis. Each sample includes a spectral plot and a detailed table summarizing the initial composition and actual labeled positions of elements, along with simulation parameters such as temperature and electron density.

2 Sample: 19088

2.1 Simulation Parameters

Parameter	**Value**
Temperature (T)	9141 K
Electron Density (n_e)	2.31e+14 $\mathrm{cm}^{-}3$

Table 1: Simulation parameters for sample 19088.

2.2 Composition Analysis

Element	**Initial Composition (%)**	**Actual Labeled Position (%)**
Al	0.00	0.00
Ar	0.00	0.00
C	4.20	6.10
Ca	15.34	3.66
Cl	2.38	0.81
Co	12.34	34.28
Cr	3.73	9.69
Fe	12.86	67.14
Mg	8.78	8.35
Mn	4.86	17.24
N	7.36	9.79
Na	9.64	7.86
Ni	4.18	11.57
O	0.00	0.00
S	1.74	4.86
Si	12.56	5.44
Ti	0.00	0.00
background	N/A	14.48
Total (%)		201.27

Table 2: Composition analysis for sample 19088.

2.3 Spectral Plot

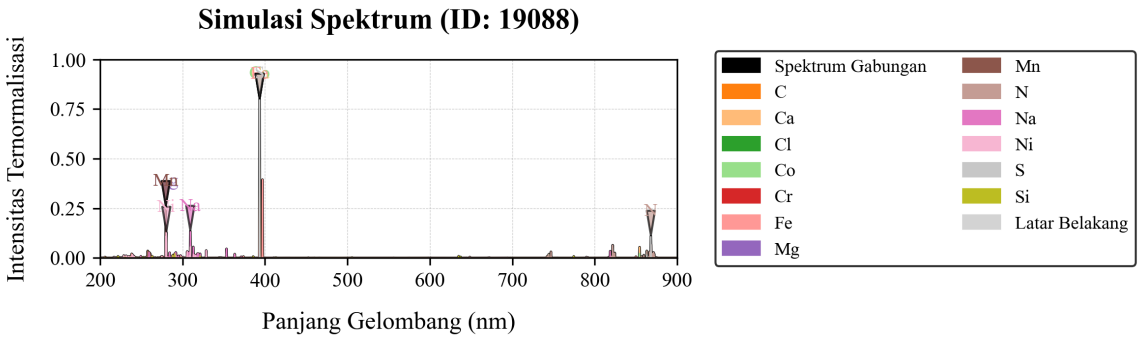

Figure 1: Spectral simulation plot for sample 19088.

3 Sample: 16764

3.1 Simulation Parameters

Parameter	Value
Temperature (T)	13081 K
Electron Density (n_e)	3.13e+15 $\mathrm{cm}^{-}3$

Table 3: Simulation parameters for sample 16764.

3.2 Composition Analysis

Element	Initial Composition (%)	Actual Labeled Position (%)
Al	10.62	11.28
Ar	6.31	17.75
C	7.26	17.90
Ca	8.19	3.66
Cl	8.37	0.85
Co	7.27	34.28
Cr	11.28	9.69
Fe	0.00	0.00
Mg	8.27	8.59
Mn	9.04	17.24
N	1.32	14.70
Na	2.78	7.86
Ni	1.59	11.55
O	4.57	7.74
S	8.48	5.52
Si	4.66	5.44
Ti	0.00	0.00
background	N/A	18.04
Total (%)		192.09

Table 4: Composition analysis for sample 16764.

3.3 Spectral Plot

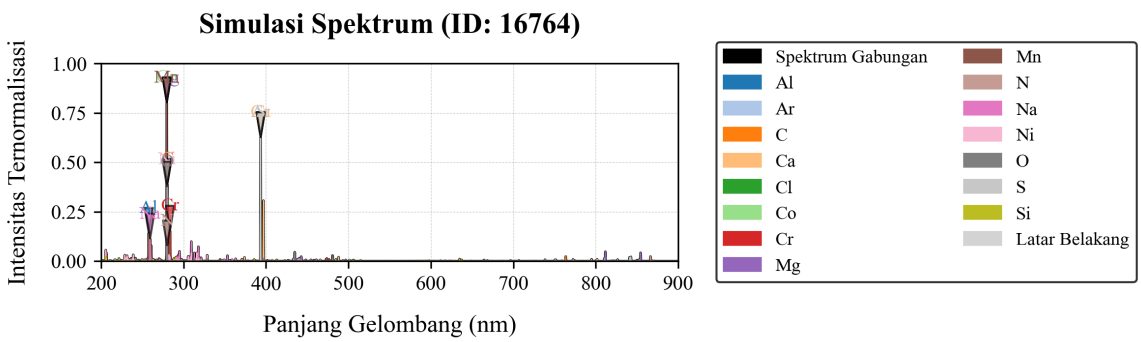

Figure 2: Spectral simulation plot for sample 16764.

4 Sample: 3467

4.1 Simulation Parameters

Parameter	Value
Temperature (T)	9242 K
Electron Density (n_e)	3.68e+14 $\mathrm{cm}^{-}3$

Table 5: Simulation parameters for sample 3467.

4.2 Composition Analysis

Element	Initial Composition (%)	Actual Labeled Position (%)
Al	5.21	10.72
Ar	0.00	0.00
C	0.00	0.00
Ca	18.30	3.66
Cl	10.98	0.85
Co	0.00	0.00
Cr	6.79	9.69
Fe	0.00	0.00
Mg	11.51	8.45
Mn	5.42	17.24
N	0.00	0.00
Na	13.93	7.86
Ni	4.20	11.62
O	10.29	4.15
S	8.89	5.40
Si	4.49	5.40
Ti	0.00	0.00
background	N/A	44.24
Total (%)		129.27

Table 6: Composition analysis for sample 3467.

4.3 Spectral Plot

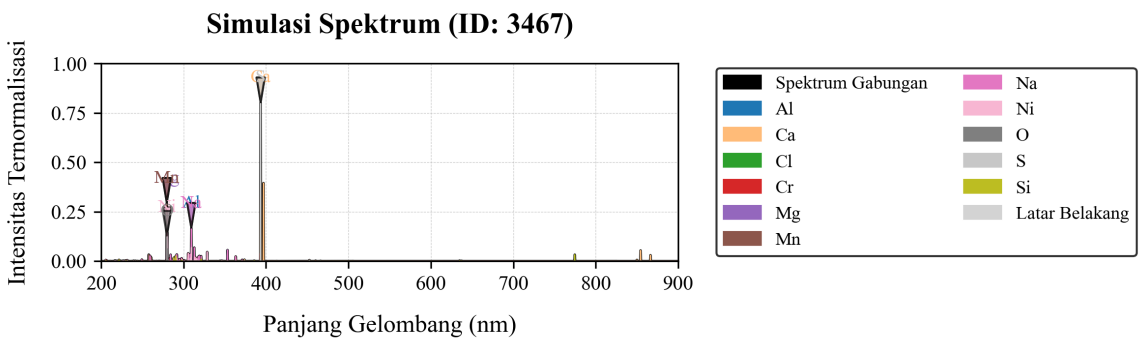

Figure 3: Spectral simulation plot for sample 3467.

5 Sample: 6423

5.1 Simulation Parameters

Parameter	Value
Temperature (T)	11162 K
Electron Density (n_e)	1.02e+15 $\mathrm{cm}^{-}3$

Table 7: Simulation parameters for sample 6423.

5.2 Composition Analysis

Element	Initial Composition (%)	Actual Labeled Position (%)
Al	0.00	0.00
Ar	0.00	0.00
C	0.00	0.00
Ca	5.87	3.66
Cl	8.92	0.85
Co	8.05	34.28
Cr	3.59	9.69
Fe	10.25	67.33
Mg	11.99	8.59
Mn	15.41	17.24
N	0.00	0.00
Na	7.24	7.86
Ni	4.60	11.57
O	0.00	0.00
S	7.84	5.57
Si	16.24	5.44
Ti	0.00	0.00
background	N/A	17.92
Total (%)		190.01

Table 8: Composition analysis for sample 6423.

5.3 Spectral Plot

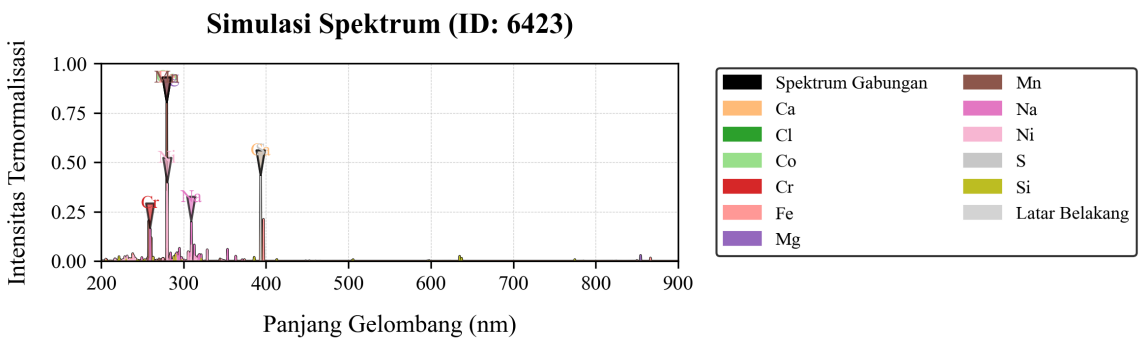

Figure 4: Spectral simulation plot for sample 6423.

6 Sample: 13548

6.1 Simulation Parameters

Parameter	Value
Temperature (T)	12071 K
Electron Density (n_e)	3.35e+14 $\mathrm{cm}^{-}3$

Table 9: Simulation parameters for sample 13548.

6.2 Composition Analysis

Element	Initial Composition (%)	Actual Labeled Position (%)
Al	11.72	11.28
Ar	0.00	0.00
C	0.00	0.00
Ca	17.72	3.66
Cl	4.47	0.85
Co	0.00	0.00
Cr	8.59	9.69
Fe	0.00	0.00
Mg	16.48	8.54
Mn	7.19	17.24
N	0.00	0.00
Na	0.92	7.69
Ni	17.46	11.57
O	0.00	0.00
S	14.71	5.52
Si	0.74	5.30
Ti	0.00	0.00
background	N/A	45.31
Total (%)		126.66

Table 10: Composition analysis for sample 13548.

6.3 Spectral Plot

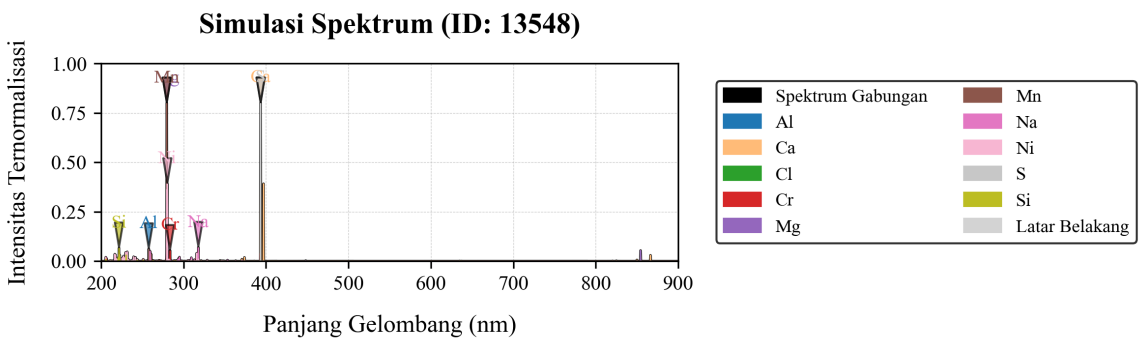

Figure 5: Spectral simulation plot for sample 13548.